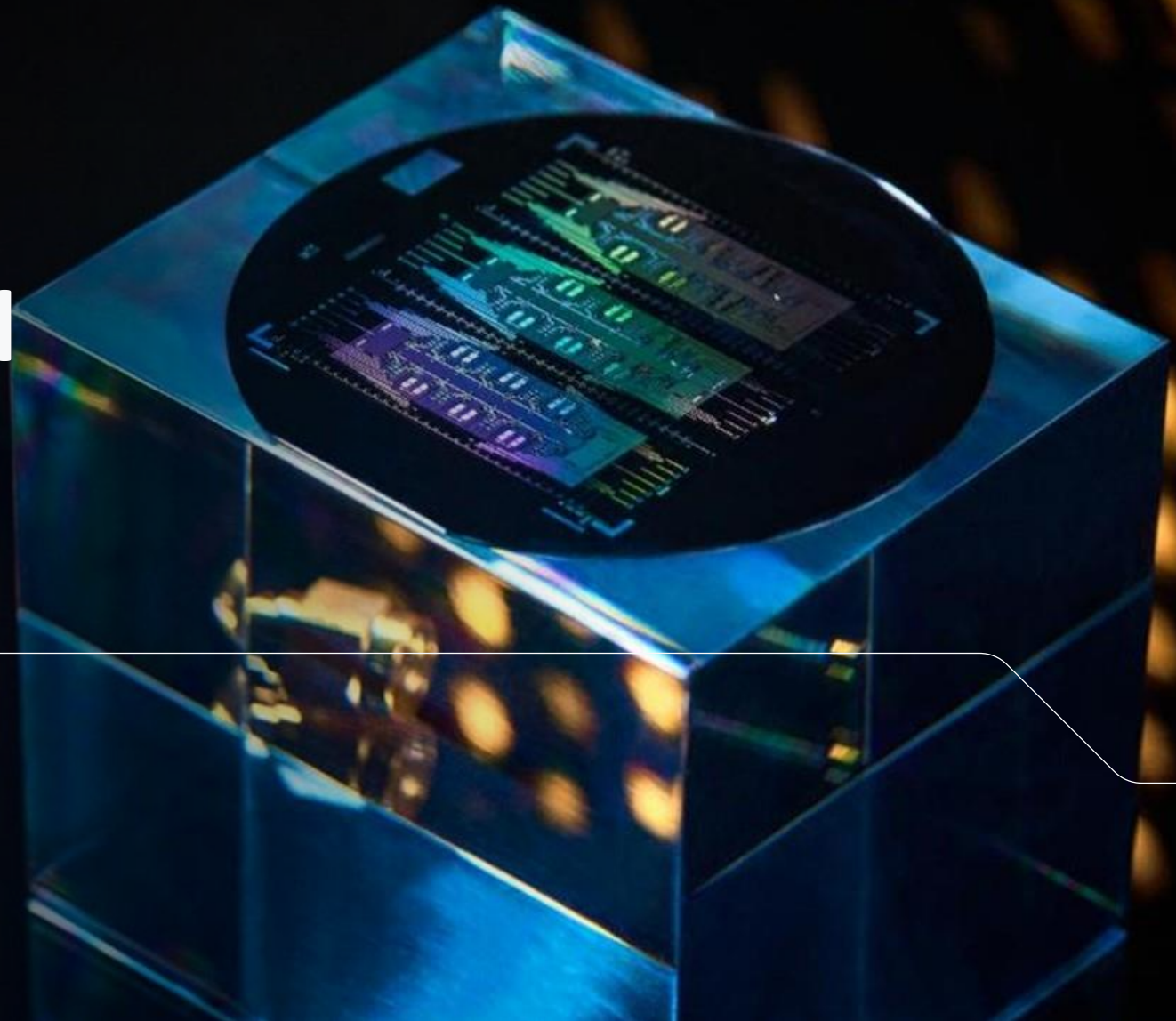
ALICE & BOB

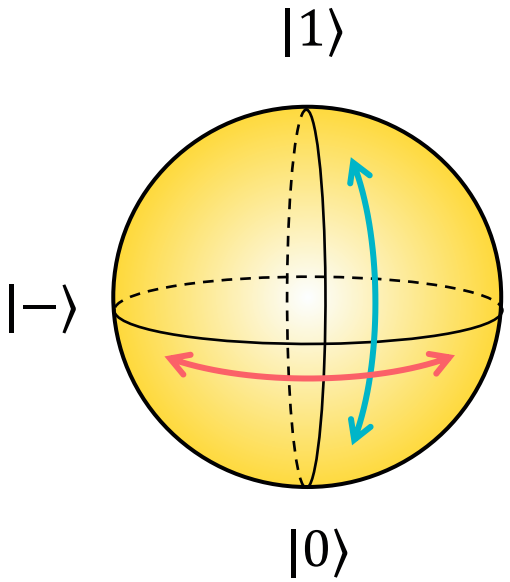
Cat qubits: Designing physical qubits for FTQC



WES ROBERTS, QUANTUM HARDWARE AND PHYSICS SPECIALIST

DUQUANTUM 2026

What are we correcting?



Bit-flip X ⟶ $\alpha|0\rangle + \beta|1\rangle \rightarrow \alpha|1\rangle + \beta|0\rangle$

Phase-flip Z ⟶ $\alpha|0\rangle + \beta|1\rangle \rightarrow \alpha|0\rangle - \beta|1\rangle$

Any single-qubit error can be represented as a linear combination of these, and their composition $Y = iXZ$.

How do we correct it?

Error correction recipe

1. Encode the qubit redundantly in a larger “code space.”
2. System is exposed to noise.
3. Perform a syndrome measurement to diagnose the error.
4. Perform a recovery operation in response.

Example: 3-qubit bit-flip code

Encoding

$$\alpha|0\rangle + \beta|1\rangle \rightarrow \alpha|000\rangle + \beta|111\rangle$$

Syndrome

$$P_0 = |000\rangle\langle 000| + |111\rangle\langle 111|$$
$$P_1 = |100\rangle\langle 100| + |011\rangle\langle 011|$$
$$P_2 = |010\rangle\langle 010| + |101\rangle\langle 101|$$
$$P_3 = |001\rangle\langle 001| + |110\rangle\langle 110|$$

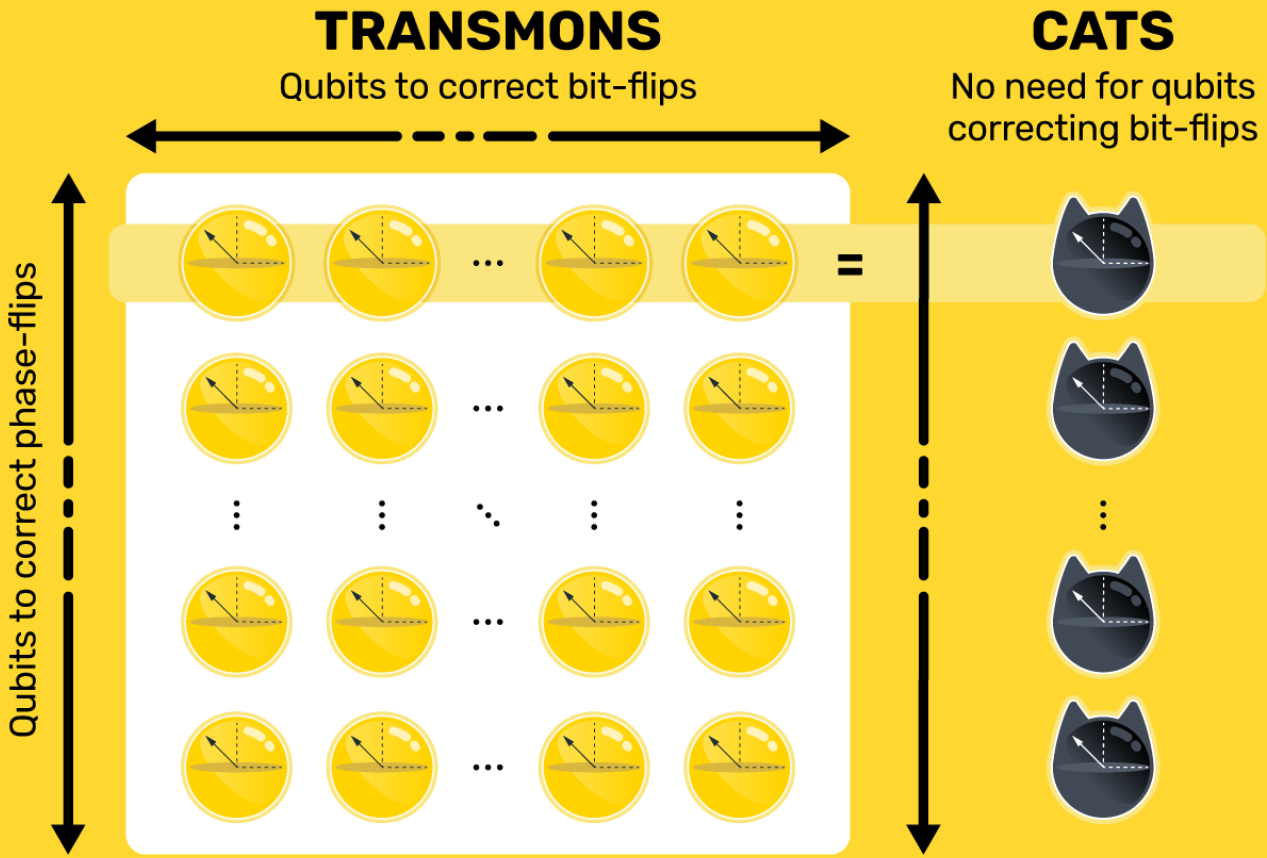
Recovery

$$I, X_1, X_2, X_3$$

See e.g. Nielsen and Chuang chapter 10

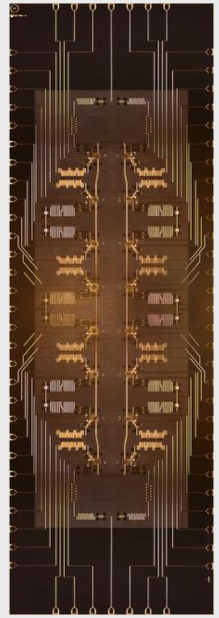
With one error suppressed in hardware, correction can be one-dimensional, only taking care of phase flips

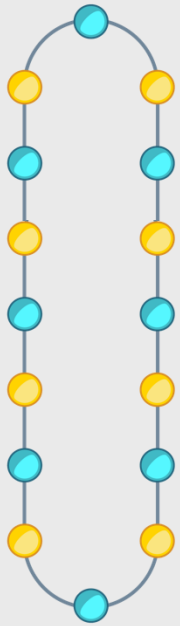


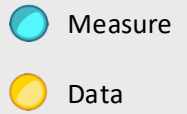


HELIUM CHIP

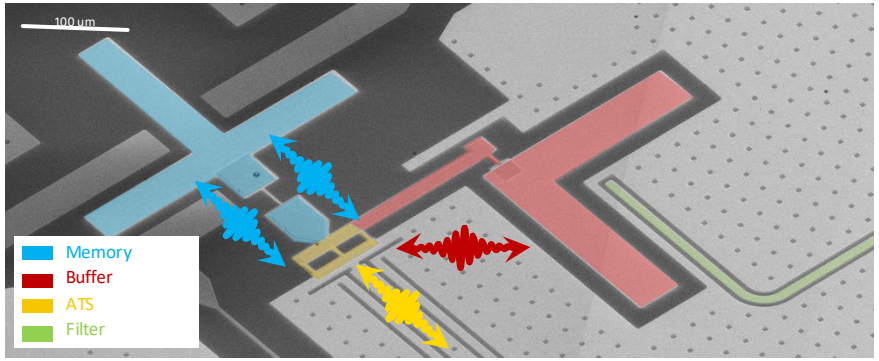
// LOGICAL QUBIT

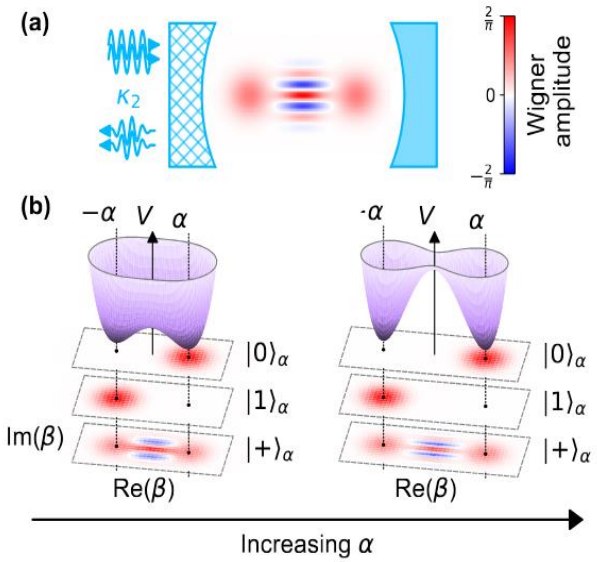






A brief introduction to **A&B's cat qubits**





A&B experimental demonstrations

2020

Founding team demonstrated first cat qubit [1]

2022

Demonstration of bit flip of multiple seconds [2]

2024

Boson 4 available on Google Cloud with bit-flip time of several minutes

2025

"Squeezed" cat qubits for improved bit-flip to phase-flip ratio [4]

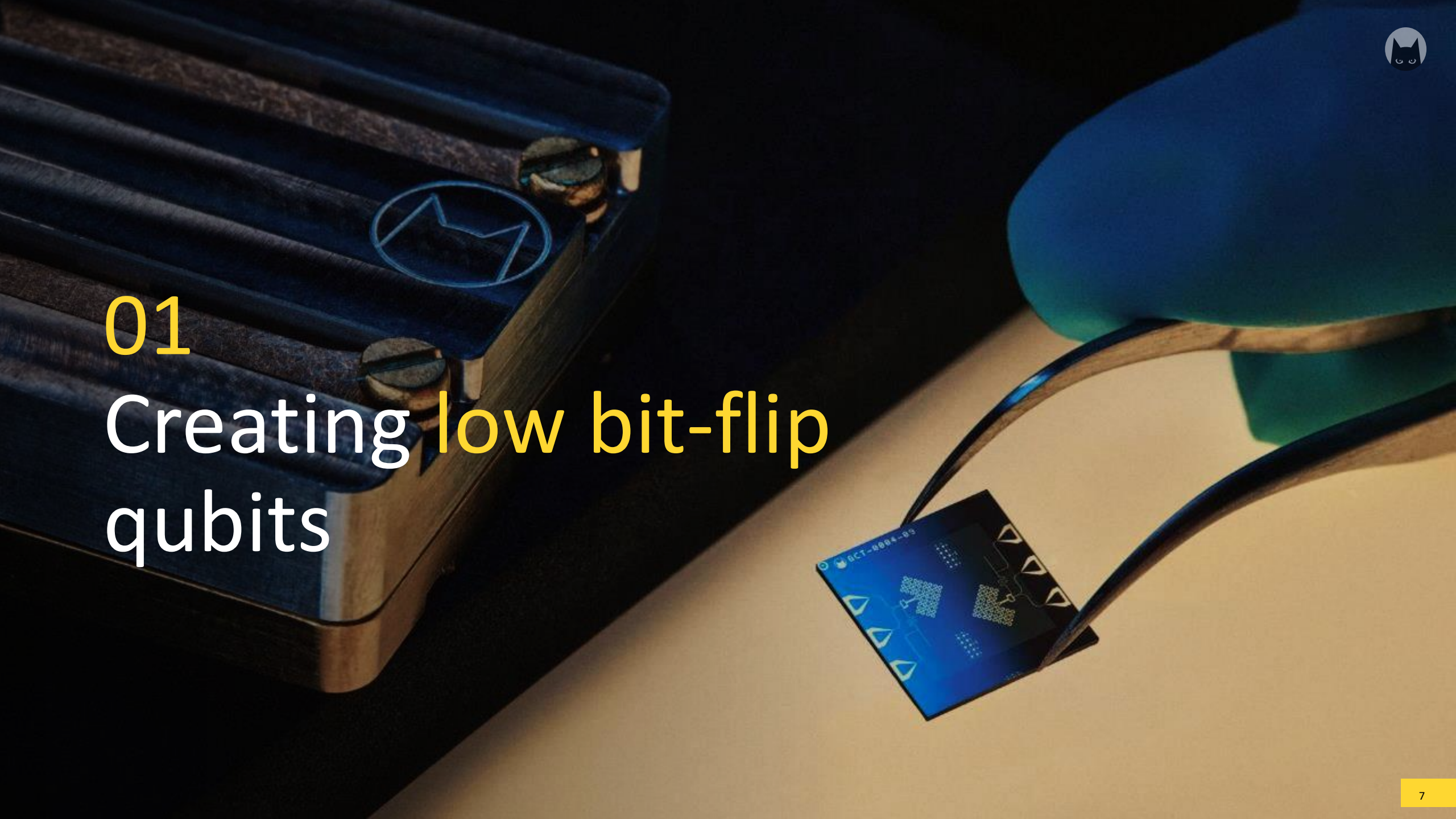
FOR THE CAT CURIOUS:

1. Lescanne, R. et al., Nat. Phys. 16, 509–513 (2020)
2. Berdou et al., PRX Quantum 4, 020350 (2023)
3. felis.alice-bob.com
4. Rousseau et al., 2502.07892 (2025)

OUR WHITE PAPER

Outline

1. Superconducting qubits
   a) Transmons vs bosonic codes
   b) Coherent states
   c) What do we mean by “noise bias” quantitatively?
2. Introduction to cat qubit design
3. Cat qubit scalability



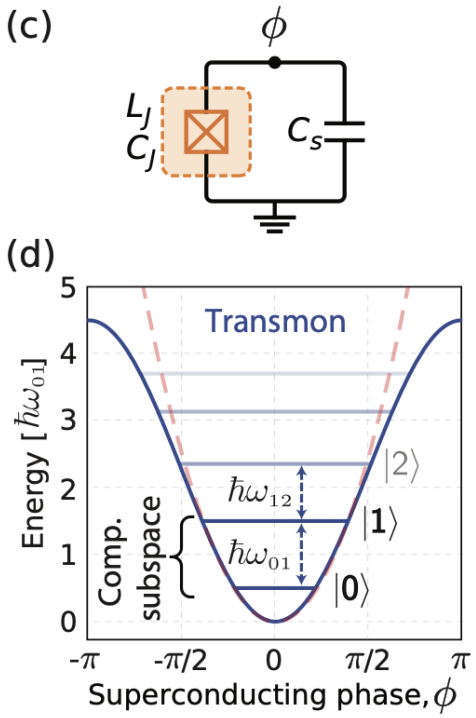
01
Creating low bit-flip qubits

Superconducting Qubits

Bosonic qubits can exploit an infinite dimensional Hilbert space, in contrast with **transmons.**

This has some convenient consequences.

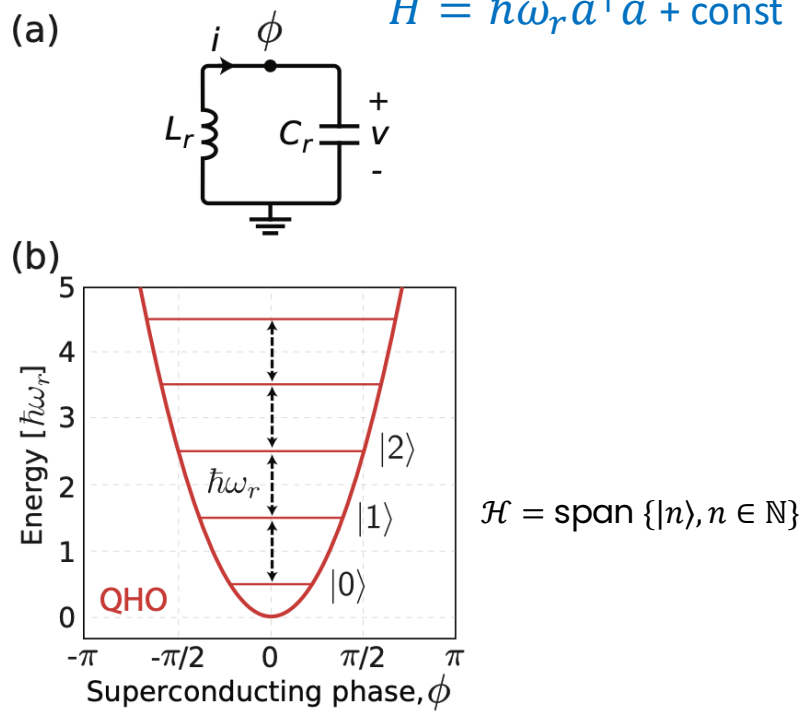
Transmon qubits



/ / QUBIT ENCODED IN THE ENERGY $\{|0\rangle, |1\rangle\}$ SUBSPACE

Bosonic qubits

$$H = \hbar\omega_r a^\dagger a + \text{const}$$



$\mathcal{H} = \text{span}\,\{|n\rangle, n \in \mathbb{N}\}$

/ / QUBIT ENCODED IN A MORE ARBITRARY SUBSPACE

FIGURES FROM P. Krantz et. al, A quantum engineer's guide to superconducting qubits, Applied Physics Reviews (2019)

Coherent states and photon loss

$$|\alpha\rangle = |0\rangle, |-\alpha\rangle = |1\rangle$$

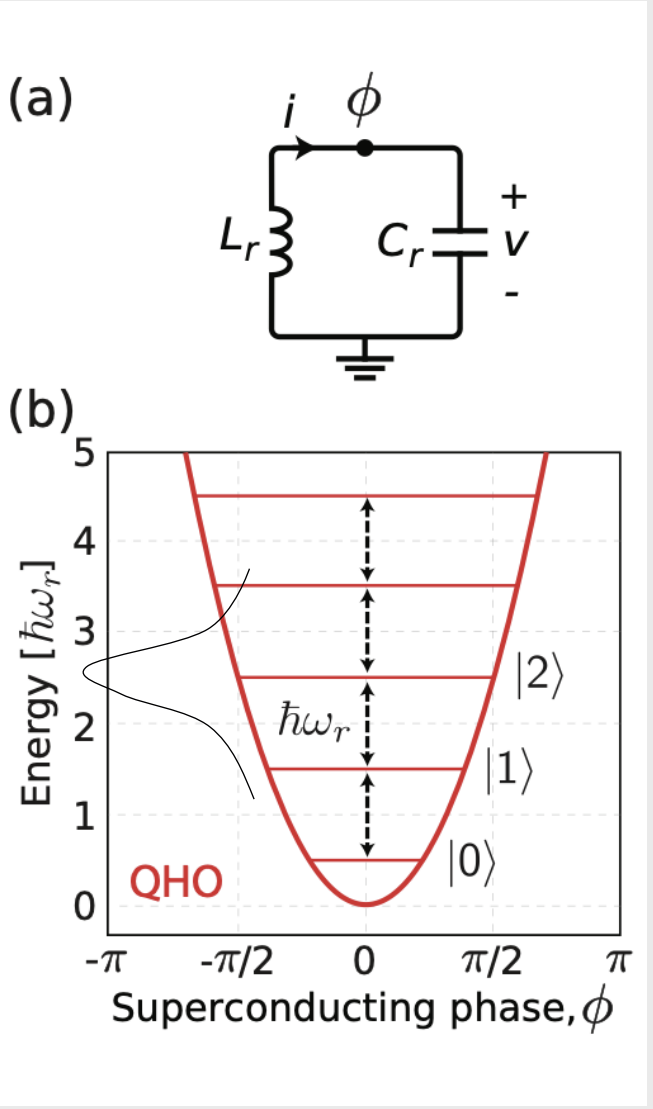
- Dominant error channel: photon loss, lowering operator a
- We can construct special states called *coherent states* with some nice properties for qubit engineering

$$a|\alpha\rangle = \alpha|\alpha\rangle$$

$$a(|\alpha\rangle + |-\alpha\rangle) = \alpha(|\alpha\rangle - |-\alpha\rangle)$$

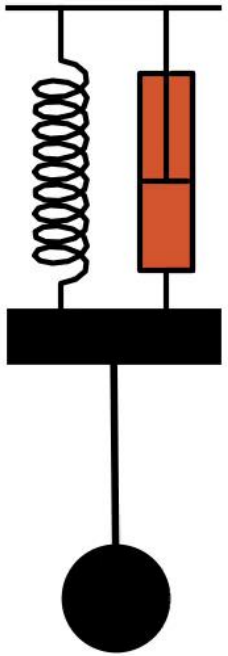
- Coherent states offer *noise bias* when loss is the dominant error:

$$\Gamma_X \sim e^{-c|\alpha|^2} \qquad \Gamma_Z \sim |\alpha|^2$$

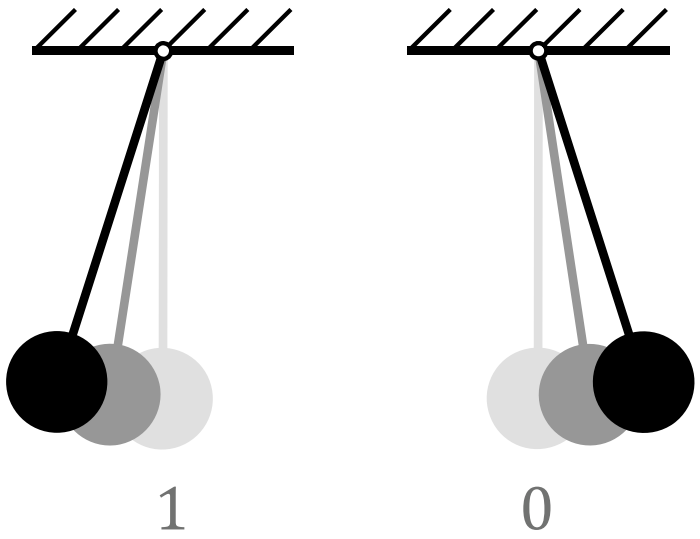


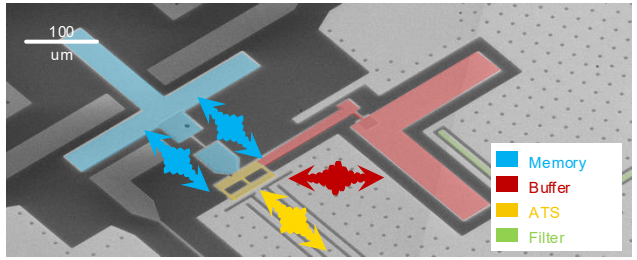
Dynamic stabilization

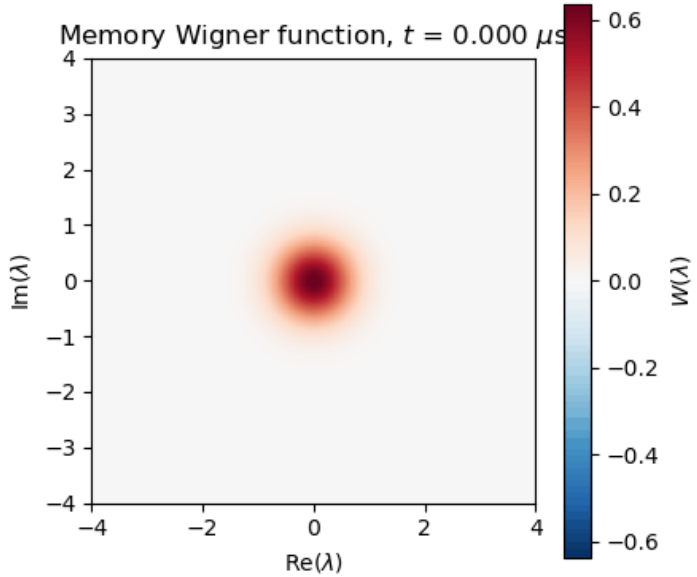
We engineer the following jump operator which stabilizes $|\pm\alpha\rangle$:



$$L = a^2 - \alpha^2$$

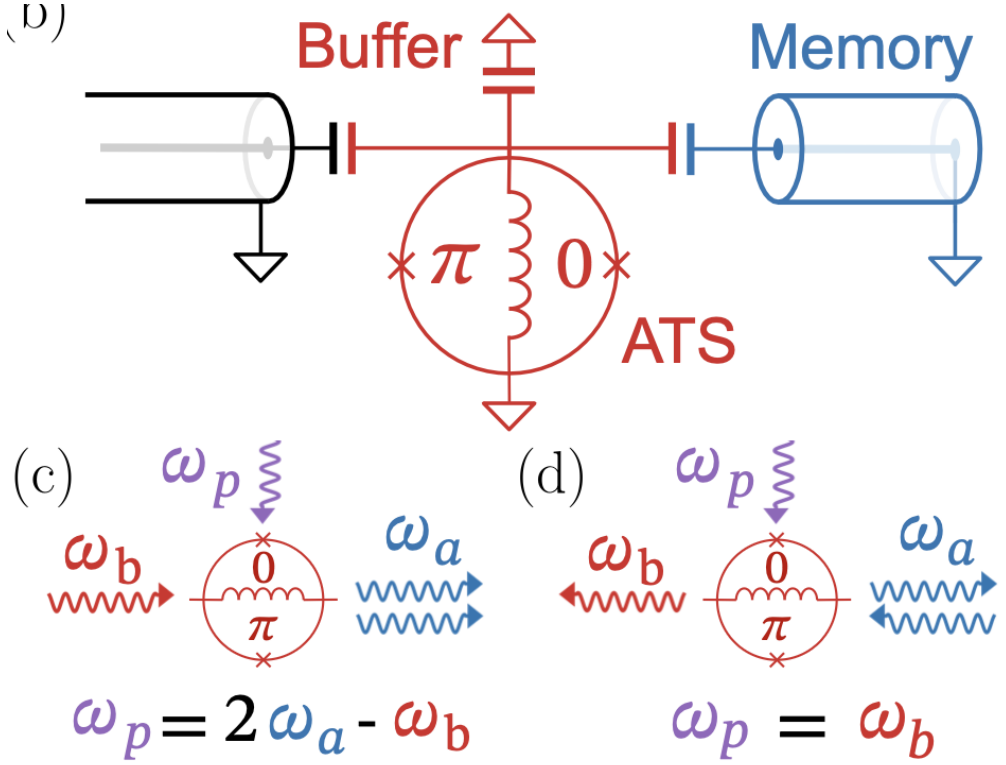




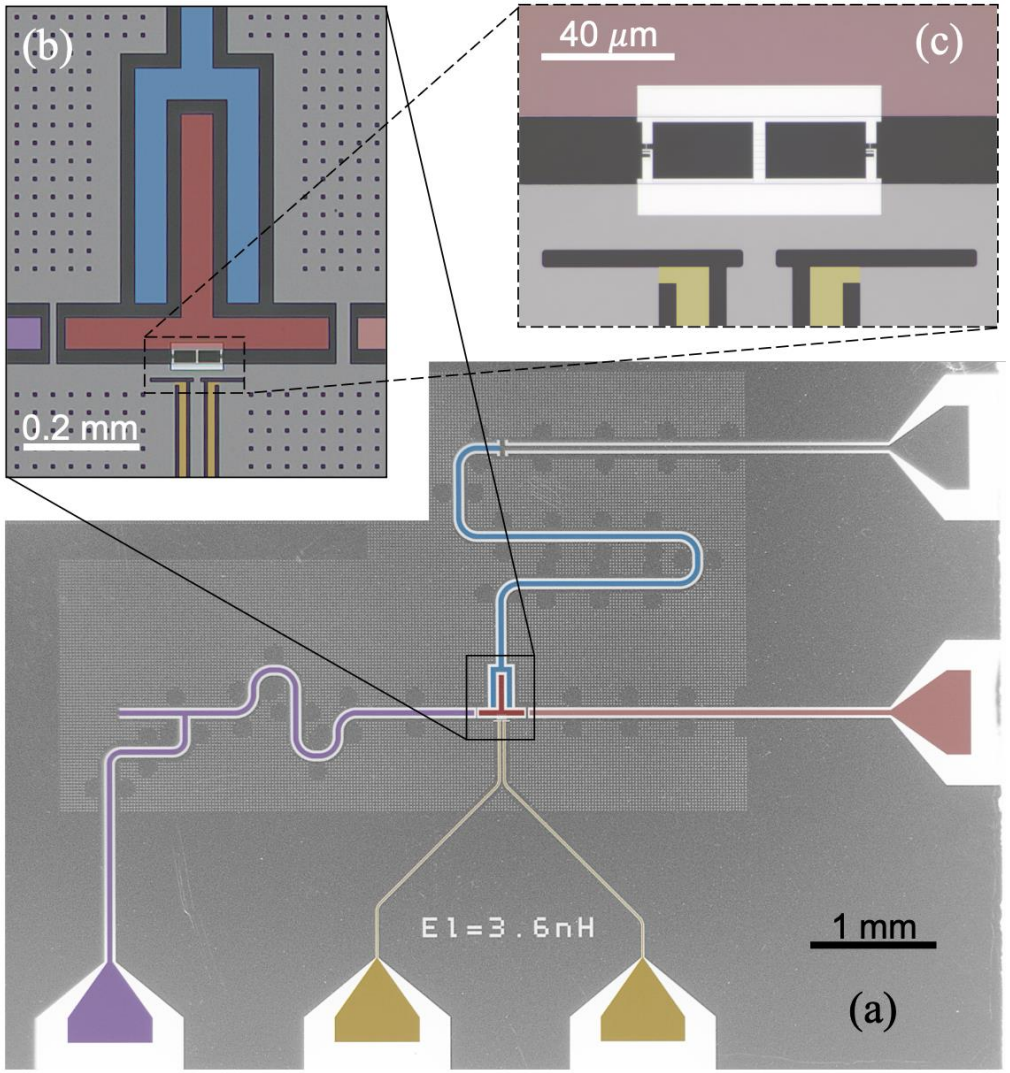


A. Marquet et al., 2023. arXiv:2307.06761

Superconducting circuit



ATS = Asymmetrically threaded SQUID
SQUID = Superconducting quantum interference device

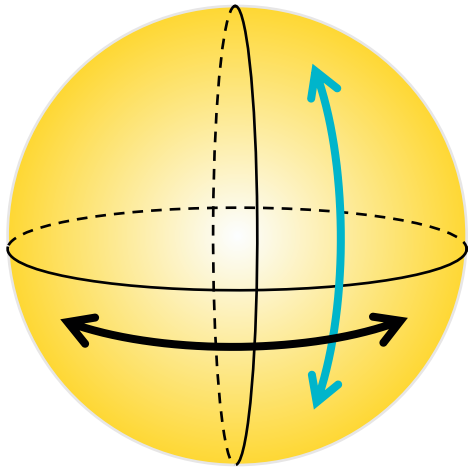


U. Réglade et al., 2024. arXiv:2307.06617v2



02
Implications for scalability

The dissipative cat qubit: a biased noise qubit



Bit-flip X

Phase-flip Z

TRANSMON QUBITS

$$T_X = T_Z = 10 - 100\ \mu s$$
$$T_Z/T_X \sim 1$$

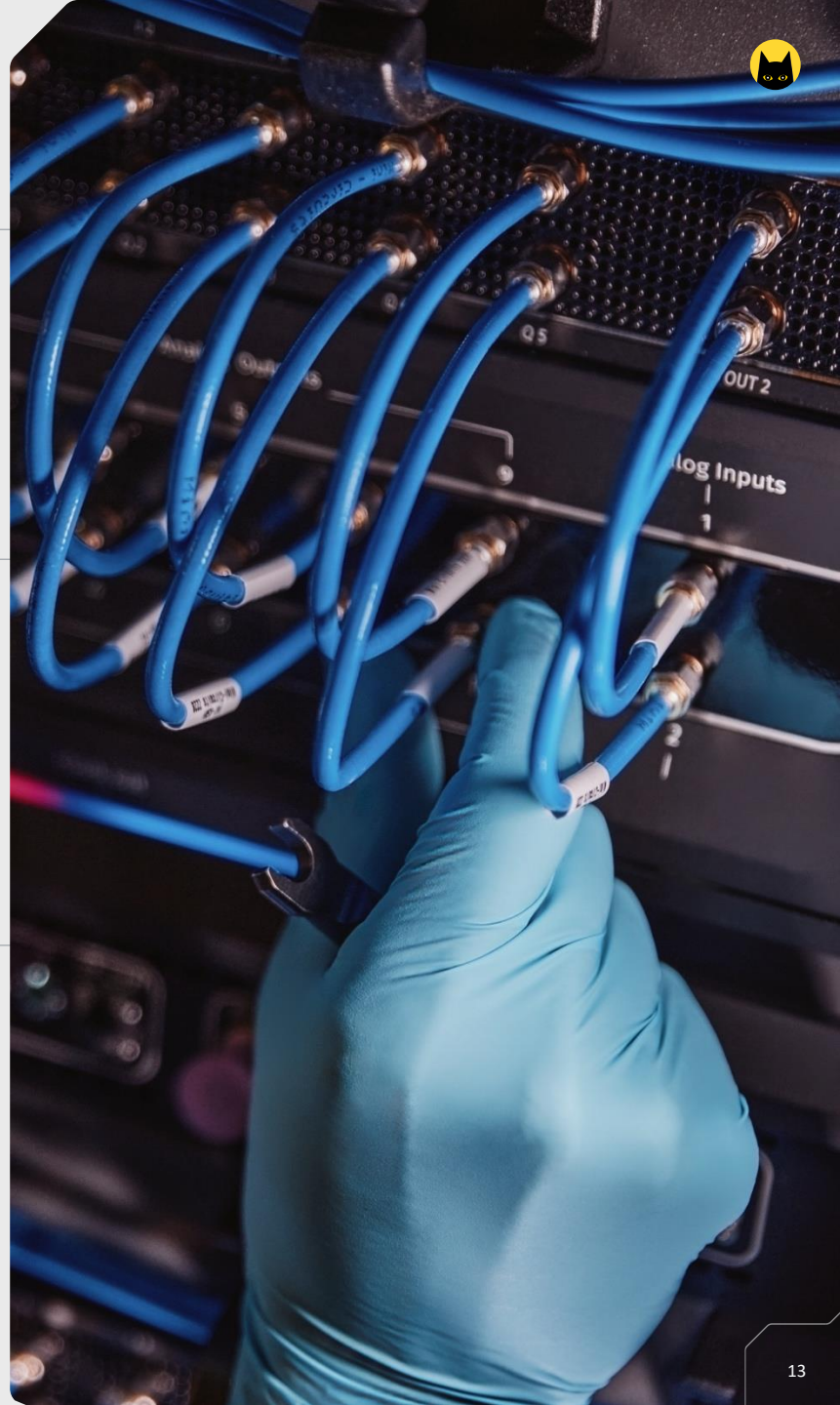
CAT QUBITS

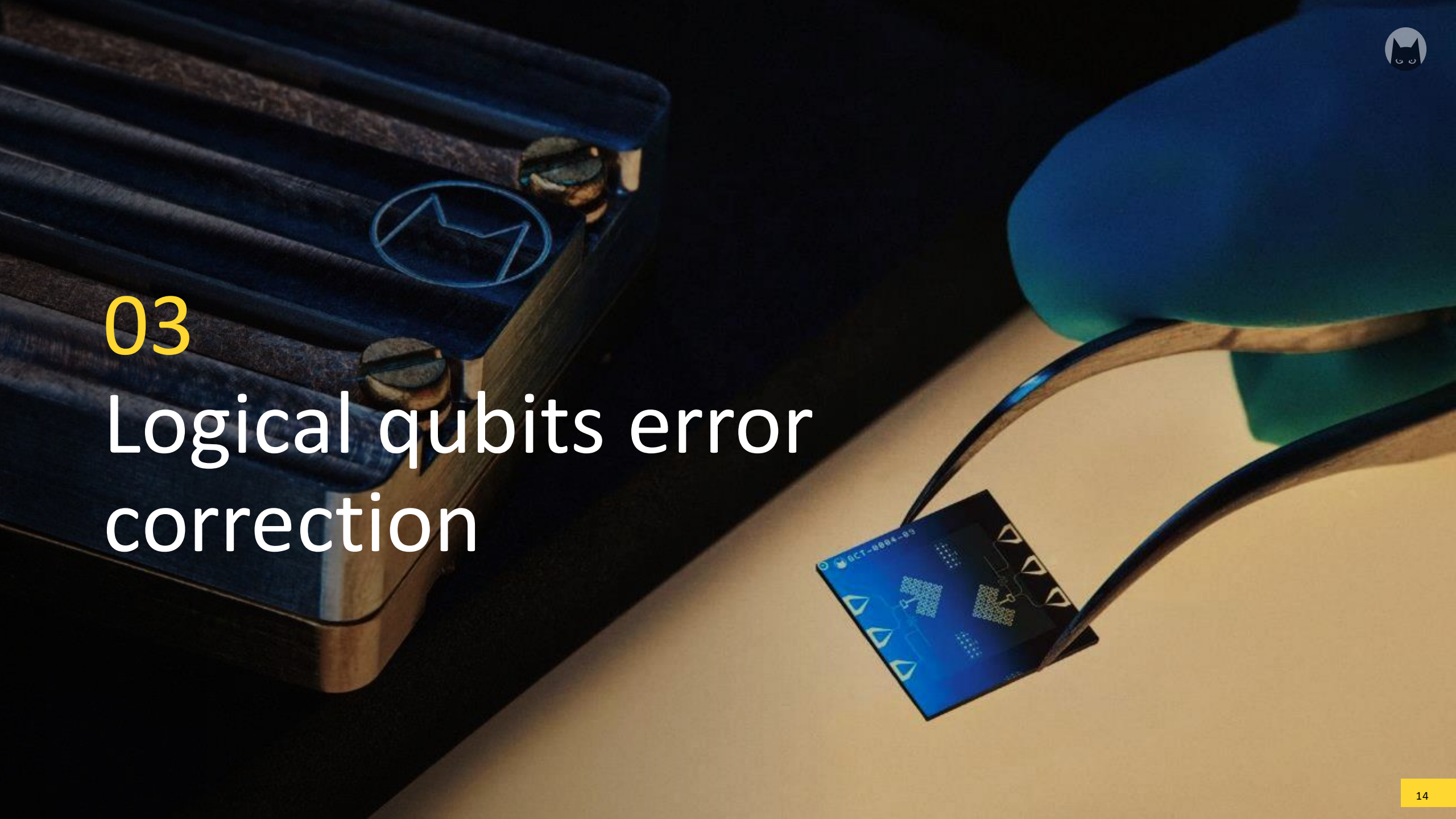
$$T_X = 10\ s \quad T_Z = 1\ \mu s$$
$$T_Z/T_X \sim 10^7$$

Bit-flip X $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \rightarrow |\psi\rangle = \alpha|1\rangle + \beta|0\rangle$

Phase-flip Z $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \rightarrow |\psi\rangle = \alpha|0\rangle - \beta|1\rangle$

Réglade, Bocquet et. al, Nature 629, 778-783 (2024)

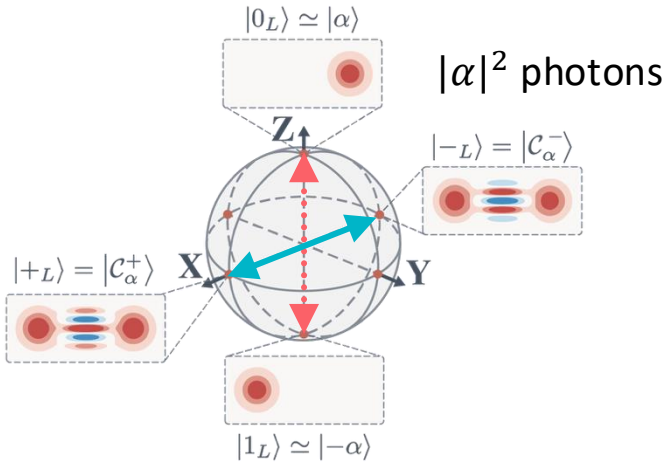




03
Logical qubits error correction

Collapsing a Dimension Bears Fruit at **Multiple Levels**

BIT FLIPS: AUTONOMOUS QEC



// STRONG NOISE BIASING:

Bit-flips are suppressed **exponentially**

$\Gamma_X \propto e^{-\gamma|\alpha|^2}$

$(\gamma \lesssim 2)$

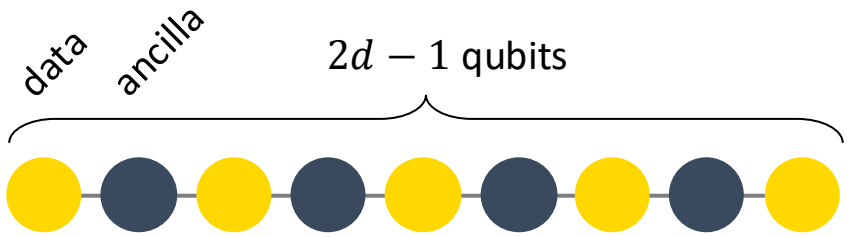
Phase-flips increase **linearly**

$\Gamma_Z = 2|\alpha|^2\kappa_1$

(single-photon loss)

Leghtas et al, Science (2015)
Lescanne et al, Nature Physics (2020)
Réglade, Bocquet et al, Nature (2024)

PHASE FLIPS: ACTIVE QEC



// 1D REPETITION CODE:

Phase-flips are suppressed **exponentially**

$\Gamma_Z \propto (p_{\mathrm{pf}}/p_{\mathrm{th}})^{\frac{d+1}{2}}$

Bit-flips increase **linearly**

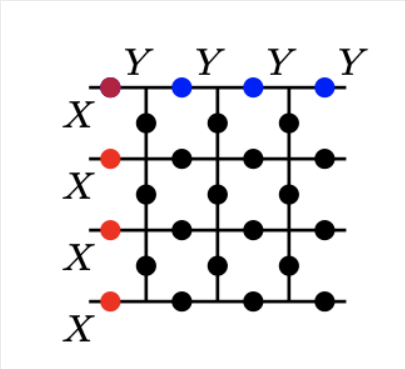
$\Gamma_X \propto d$

Guillaud & Mirrahimi, PRX (2019)
C. Chamberland et al, PRX Quantum (2022)
AWS Quantum, Nature (2025)

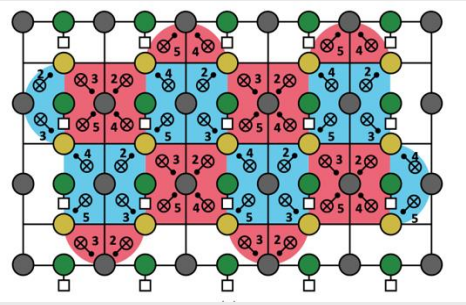
Fully protected logical qubit?

Include some bit-flip error correction capability ? → bias-tailored codes

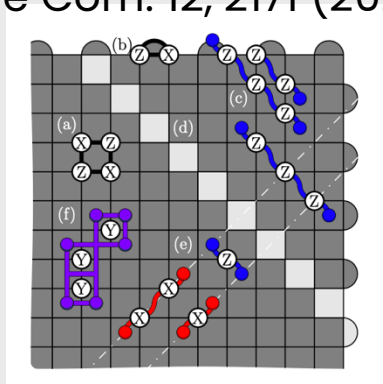
D. Tuckett *et al,* PRL 120,0505 5 (2018)



C. Chamberland *et al,*
PRX Quantum 3, 010329 (2022)

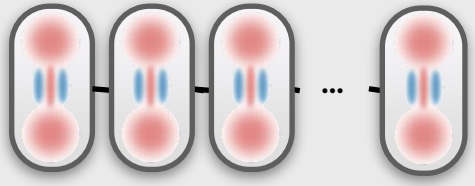


J. Pablo Bonilla Ataides *et al,*
Nature Com. 12, 2171 (2021)



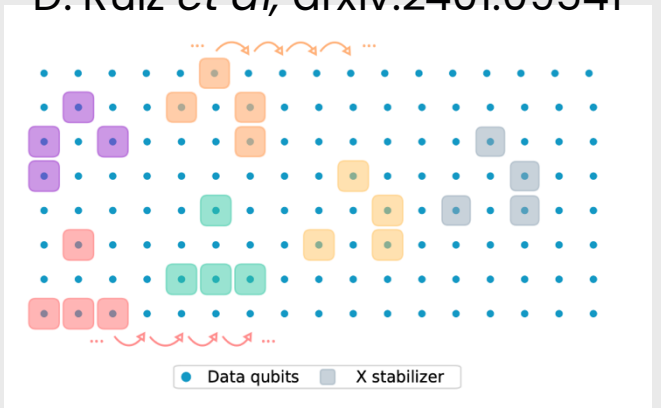
$\overline{n} = |\alpha|^2$ **large enough to handle bit-flips alone ?** → phase-flip codes (« classical » codes)

JG and M. Mirrahimi, Phys. Rev. X 9, 041053 (2019)
C. Chamberland et al, PRX Quantum 3, 010329 (2022)



H. Putterman et al, arxiv:2409.13025

D. Ruiz *et al,* arxiv:2401.09541





04

Questions?



01
Quantum error correction: a recap

Phase flip error correction

1D repetition code with n cat-qubits



$$|+\rangle_L = |++\cdots+\rangle$$
$$|-\rangle_L = |--\cdots-\rangle$$

$$Z_L : |++\cdots+\rangle \rightarrow |--\cdots-\rangle = \otimes_i Z_i$$

$$X_L : \frac{|++\cdots+\rangle + |--\cdots-\rangle}{\sqrt{2}} \rightarrow \frac{|++\cdots+\rangle - |--\cdots-\rangle}{\sqrt{2}}$$
$$= \text{any } X_i$$

Stabilizer : $X_i X_{i+1}$ measured using b.p. CX gates

- Qubit, stable (rare bit-flips)
  coherent (rare phase-flips)
- Initialize and measure $|0/1\rangle$, $|+/-\rangle$
- Physical bias-preserving gates
- Logical universal set of gate

} Logical qubit in **1D** array
Bacon-Shor like

$P_{bf} \propto e^{-\alpha^2}$	$P_{bf} \propto n$
$P_{pf} \propto \alpha^2$	$P_{pf} \propto e^{-n}$
Auto. EC	1D ECC



03 Master equation approach

Density operators: adding classical probability to quantum states

What if we don't know the precise state of a quantum system, but rather that it belongs to a *classical* distribution of *quantum* states?

$$\langle A \rangle = \sum_i p_i \langle i|A|i\rangle = \sum_i p_i \langle i| \sum_n |n\rangle\langle n|\, A|i\rangle = \sum_n \sum_i \langle n|A(p_i|i\rangle\langle i|)|n\rangle = \mathrm{Tr}\, A\rho$$

Here $\rho = \sum_i p_i |i\rangle\langle i|$ is the density operator. It contains everything there is to know about the quantum state, including our classical ignorance.

Because it must remain a "good" quantum state at all times, ρ evolves in a special way:

$$\frac{d\rho}{dt} = -i[H,\rho] + L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}$$

unitary jump operator

$$\frac{d\rho}{dt} = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}$$

Dark states and stabilization

STATIONARY STATES OF THE MASTER EQUATION

When we prepare in an arbitrary state, the dynamics ”cool” it into a dark state.

These are states where $L|\psi\rangle = 0$.

They are stabilized states, so that $\frac{d\rho}{dt} = 0$.

ENGINEERING THE DYNAMICS

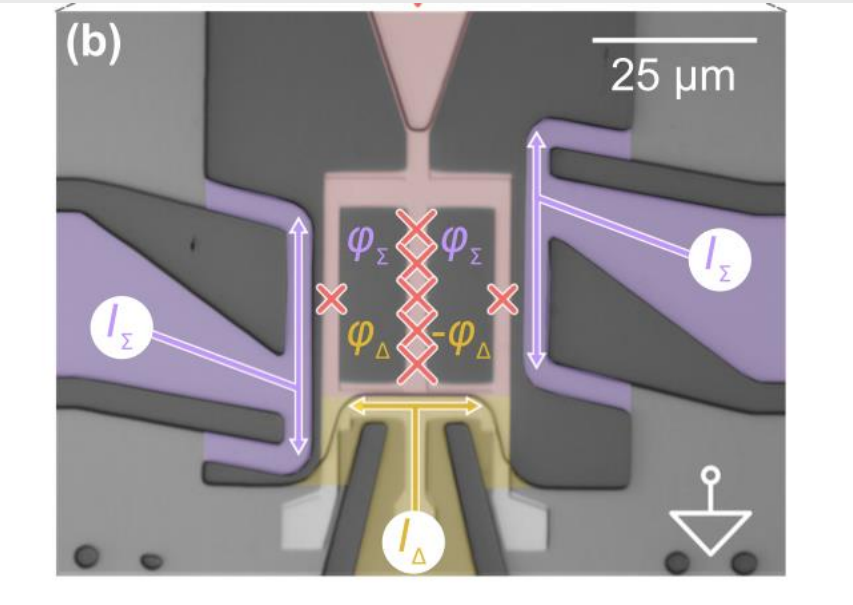
We want to engineer special dynamics (through our choice of L) such that the dark states are $|\pm\alpha\rangle$.

This will act as our “code space” for autonomous bit-flip error correction!

For a superconducting resonator, photon loss $L = a$ is the dominant error channel.



ATS Setup



R. Lescanne, et al. (2019) arXiv:1907.11729

Circuit Hamiltonian

$$H_{circuit} = \omega_a a^\dagger a + \omega_b b^\dagger b - 2E_J \cos(\phi_\Sigma)\cos(\boldsymbol{\phi} + \phi_\Delta)$$

$$\boldsymbol{\phi} = \phi_a(a + a^\dagger) + \phi_b(b + b^\dagger)$$

We wish to apply a parametric pump to the ATS (through $\boldsymbol{\phi}$) to obtain:

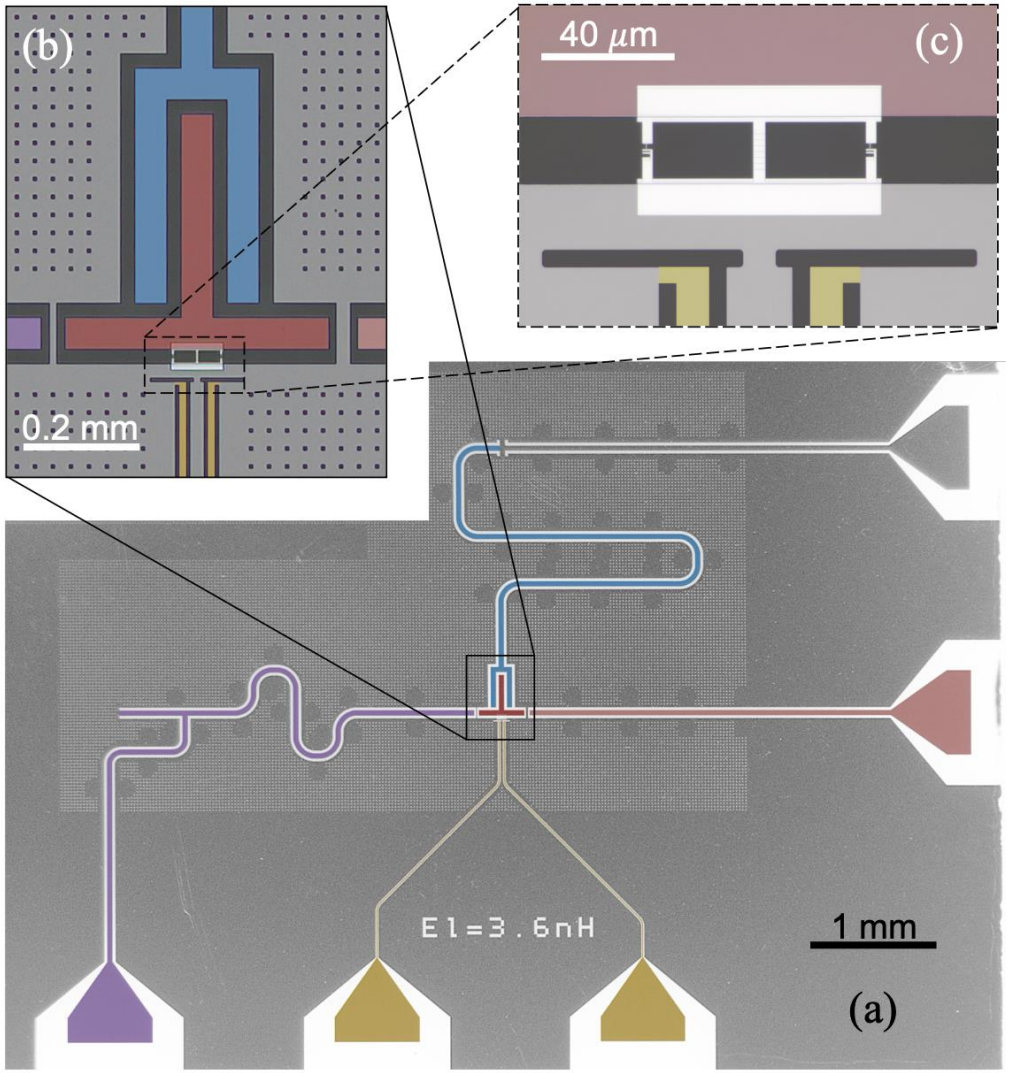
$$H = \omega_a a^\dagger a + \omega_b b^\dagger b + g_2 a^{\dagger 2} b - \epsilon_d b^\dagger + h.c.$$

Two-photon exchange with buffer

Buffer drive

Buffer is lossy: $L = \sqrt{\kappa_2} b$

U. Réglade et al., 2024. arXiv:2307.06617v2



Stabilized dynamics: semiclassical trajectories

Why this Hamiltonian? Because it stabilizes the cat states.

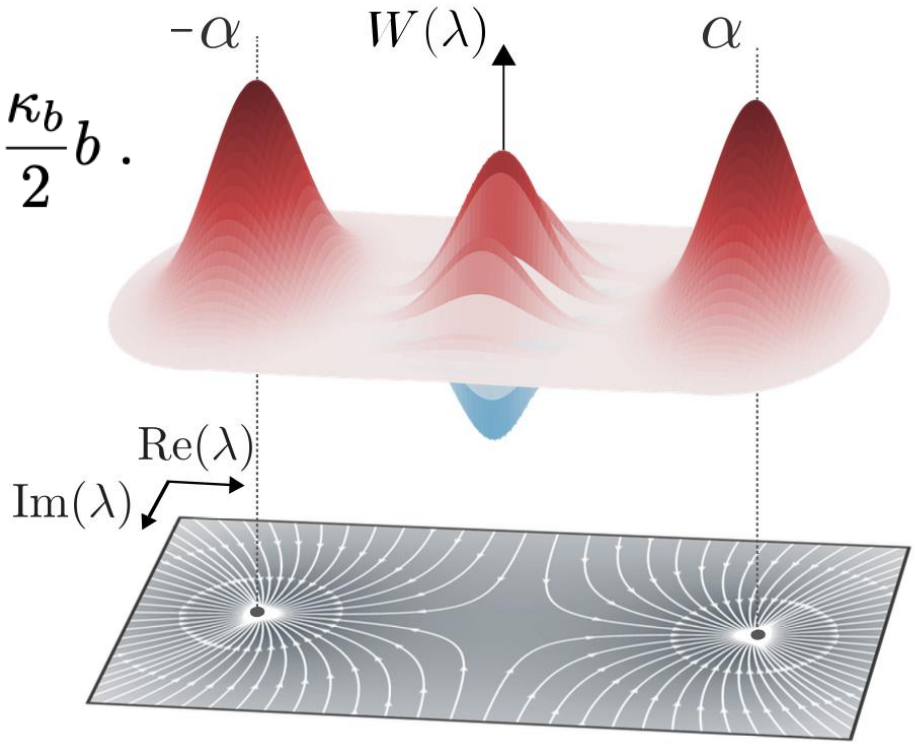
$$H = g_2^*(a^2 - \alpha^2)^\dagger b + h.c. \quad \text{where } \epsilon_d = \alpha^2 g_2^*$$

Semiclassical analysis of dynamics gives:

$$\dot{a} = -2ig_2 a^* b$$
$$\dot{b} = -ig_2^*(a^2 - \alpha^2) - \frac{\kappa_b}{2} b \,.$$

The steady states of this dynamics are $(\mathrm{a}, \mathrm{b}) = (\pm\alpha, 0)$

Equivalently: $L = a^2 - \alpha^2 \longrightarrow \frac{d\rho}{dt} = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}$



U. Réglade et al., 2024. arXiv:2307.06617v2

Transmon-free readout: memory bit measurement

- Intuitive picture:
  1. displace the memory slightly
  2. Image of displacement is imprinted on the buffer
  3. Collect info about memory bit from buffer

- Apply a memory drive of $H_Z = \epsilon_Z a^\dagger + \epsilon_Z^* a$

- Semiclassical dynamics near $a = \pm\alpha$ now gives

$$b = \langle b \rangle = \mp \frac{\epsilon_Z}{2\alpha^* g_2}$$

- Increasing ϵ_Z increases readout contrast, but also introduces spurious terms

Rotating wave approximation

- We drive the system at frequency ω_p: $-2E_J\epsilon_p \cos(\omega_p t) = E_J\epsilon_p\left(e^{i\omega_p t} + e^{-i\omega_p t}\right)$.
- Working in the interaction picture, our bosonic operators evolve as $a(t) = ae^{-i\omega_a t}$.
- Pumping at a particular frequency allows us to select certain resonant modes. Fast modes are discarded (average to zero on the characteristic timescale of the problem).
- For example, pumping at $\omega_p = \omega_b$ resonantly selects the term b because
  - $\left(e^{i\omega_b t} + e^{-i\omega_b t}\right)be^{-i\omega_b t} = be^0 + be^{-2i\omega_b t} \to b$
- Likewise:
  - $\left(e^{i\omega_b t} + e^{-i\omega_b t}\right)a^\dagger e^{i\omega_a t} a e^{-i\omega_a t} be^{-i\omega_b t} = a^\dagger a\, be^0 + a^\dagger a\, be^{-2i\omega_b t} \to a^\dagger ab$

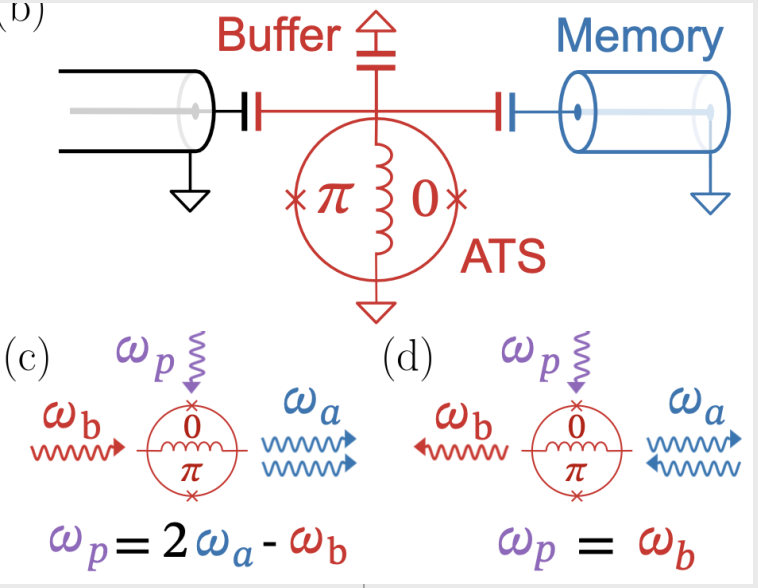
Parametric pump for 2-photon dissipation

$$H = \omega_a a^\dagger a + \omega_b b^\dagger b - 2E_J \cos(\phi_\Sigma)\cos(\boldsymbol{\phi} + \phi_\Delta) \rightarrow \omega_a a^\dagger a + \omega_b b^\dagger b + g_2 a^{\dagger 2} b - \epsilon_d b^\dagger + h.c.$$

- This Hamiltonian is stabilized by applying a flux bias to the ATS of $(\phi_\Sigma, \phi_\Delta) = \left(-\pi/2 + \epsilon_p \cos(\omega_p\, t)\,, -\pi/2\right)$
- In the regime where ϵ_p ≪ 1 we have

$$U(\phi) \approx -2E_J\epsilon_p cos(\omega_p t)\left(\phi - \frac{1}{6}\phi^3\right)$$

- Pumping at $\omega_p = 2\omega_a - \omega_b$ and applying a rotating wave approximation selects the term:



$$U(\phi) = g_2\left(a^2 b^\dagger + a^{\dagger 2} b\right)$$

$$g_2 = \frac{1}{2} E_J \epsilon_p \phi_a^2 \phi_b$$